Research Statement

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I specialize in moral and political philosophy, with particular interests in normative powers, the ethical significance of interpersonal relationships, the attractions and limits of individualist approaches to morality, and the ethics of AI. Much of my work seeks to do justice to the idea that our duties are both *explained by our relationships* and *justified individualistically*, from one person to another. I maintain related research programs in logic and philosophy of action and language. I am interested in how formal models of moral language and unfair AI can reveal and obscure injustices, and my research shows that many pressing ethical and political questions, for example regarding wronging and discrimination, are bound up with questions about the nature of causal explanation. As this summary might suggest, I am a generalist at heart.

1. Reasons by Request

By issuing requests of one another, we wield what is, on reflection, a remarkable power: to create reasons at will. For example, you are not required to stop and aid every stranger you encounter. Seeing a lost tourist, you can permissibly decide not to help, thinking to yourself, "I'll help the next time I see someone in their shoes." But this thought rings hollow when the lost stranger *requests* your help: the stranger's request gives you a reason, and you must now cite a reason for not helping or else provide assistance.

The central puzzle of requests is to explain how this is possible. In **How Requests Create Reasons** (*Philosophy & Phenomenological Research*, 2026), I argue that we have reasons to grant requests when doing so improves our compliance with duties. For example, we have a duty of mutual aid, but because this duty gives us the discretion to decide exactly when and how we will help others, we are liable to fall short of it, by repeatedly postponing compliance. When a lost stranger asks you for help, their request gives you a reason *here* and *now* to help *this person*. The concrete nature of the request makes it harder to keep postponing your duty of mutual aid. Requests create reasons because they thereby help us to comply better with our flexible moral duties.

I next consider how the values served by the power to request constrain its permissible use. In **Why Not Ask?** (R&R, *Ethics*), I argue that some requests are wrongful *because they create reasons* and thereby put the requestee in a double-bind: if the requestee acts as requested, they pay the practical cost of doing so; if they do not, they pay a moral cost, in the form of a reason for regret. While this double-bind does not always make requests wrongful, I show that in more serious cases it can be intrusive, manipulative, and coercive.

Having argued that the power to request is justified by the value of improved compliance with duties, I address a foundational puzzle: How can this power be explained *by its value*? After all, it would be wishful thinking to infer that we have the ability to fly, simply because it would be valuable. In **What Justifies Normative Powers?**, I argue that the power to create reasons by requesting can be justified individualistically, by comparing its value *for each person:* a person possesses this power when it is generally better for *requesters* in these circumstances to *possess* the power than it is for *requestees* to *not be subject* to it. After extending my view to commands and promises, I argue that it rests on a familiar and widely-accepted form of ethical explanation, which justifies moral principles by considering what the world would be like, were those principles generally accepted.

2. Valuable Relationships

I argue that a range of moral and political phenomena, including grace, reciprocity, corruption, and opportunism, are best understood in terms of the way they affirm or set back relationships that matter to us in themselves. In **Giving Grace** (invited to *Oxford Studies in Normative Ethics*), I argue that we often have duties to *give grace:* to forgo fitting blame out of regard for those who have wronged us. Such duties are puzzling, because those who have wronged us do not deserve grace—instead, they deserve blame. To dissolve this puzzle, I observe that blame also leaves scar tissue: a relationship is damaged by repeated blame, even when it is resolved on each occasion, since blame involves feelings of disappointment and resentment, the suspension of trust, and the withdrawal of good-will. I propose that our duties to give grace are justified by the fact that we share interests with those who have wronged us in the value of our relationship. Because I appeal to the *cumulative effects* of repeated blame on a relationship, my view respects wrongdoers' interests in receiving grace over time, as well as victims' discretion to decide on each occasion whether and how they give grace.

Reciprocity is a basic feature of many of our most valuable relationships and communities. It finds expression in friendly exchanges of gifts, the shared responsibilities of a family making dinner together, and civil engagement in public discourse. In **Two Kinds of Reciprocity** (under review), I argue that paradigmatic forms of reciprocity, including reciprocity in kind and paying it forward, will remain puzzling so long as we fail to distinguish *material reciprocity*, which is realized by matching benefits received with benefits given, from *relational reciprocity*, which is realized by doing one's fair share to contribute to a valuable relationship. While material and relational reciprocity can be mutually supporting, I argue that they are distinct, and that the distinction between them sheds new light on a range of practical issues, including political obligation, gratitude, the morality of markets, and our reasons to care about future generations. An in-progress paper further develops this research program. In **Betting on Misfortune**, Johann Frick and I argue that prediction markets facilitate a morally objectionable form of opportunism, insofar as they give participants reasons to hope for others' misfortune, thereby limiting their ability to relate to others with compassion.

3. Individualism and Its Limits

According to *Individualism*, our duties are explained by the reasons that individuals have to want their own lives to go one way or another, without adding these reasons together. Versions of this idea date back to the Enlightenment, but today it is exemplified by Thomson's rights theory and by Scanlon's contractualism. Individualism has two major attractions. First, as Léa Bourguignon and I explain (**How to Count Sore Throats**, *Analysis*, 2025), individualism nicely limits moral aggregation. For example, one ought to save a life rather than prevent any number of headaches, since the reasons to prevent headaches do not aggregate. With Léa Bourguignon and Jonas Harney (**Individualism and Its Limits**, R&R, *Noûs*), I revive a long-standing challenge for individualism: it cannot explain why we should save five lives rather than one. I conclude that individualism is best seen as an account of a non-aggregative subdomain of morality concerned not with what is *permissible*, but with what we *owe to one another*.

Indeed, the second major aspiration of individualism is to explain why some acts are not just wrong but also *wrongings* against particular victims, i.e. violations of duties owed to others. E.g., breaking a promise standardly wrongs the promisee. But it has proven difficult to spell out the individualist account of wronging in a precise and plausible way. With Russell McIntosh (**Wronging in the Image of Causation**), I answer this challenge by developing a simple theory of wrongings, which draws a deep parallel between causal and moral explanation. Mackie said that a *cause* of an event is simply a necessary part of a set of conditions that suffice for that event to occur, given the laws of nature. E.g., striking a match causes the fire, because together with other conditions—the presence of oxygen, and so on—the match-striking suffices for the fire to occur, given the laws of nature. We argue that an act *wrongs* an individual when their interests are a necessary part of a set of conditions that suffice for the act to be impermissible, given the laws of morality. E.g., if I break a promise to you and thereby make John lose a bet, I wrong you and owe you apology and compensation, but I have no such duties to John. This is because your interests alone suffice to make my act impermissible, but John's do not.

4. Causal Models of Discrimination

According to U.S. and U.K. law, direct gender discrimination occurs when a person's gender *causes* their worse treatment. But what does it mean to *only* vary gender and hold other features fixed, given that gender is bound up with so many other attributes? Suppose Maya was a Girl Scout, went to Barnard, became a nun, and is wearing a dress. To *only* vary gender, should we imagine a man with these features? Or a former Boy Scout who went to Columbia, became a priest, and is wearing robes? Which features should remain unchanged, and which would need to vary (and on what grounds)? A central problem in conceptualizing discrimination is to explain how it makes sense to think of protected attributes like gender and race as causes, given that these attributes are embedded in a nexus of social factors that resist isolated treatment.

Faced with this problem, some give up on modeling gender and race as social kinds, rather than strictly biological ones. Others suggest that we give up on traditional causal models for discrimination, which require the ability to isolate causes from other attributes. With Frederick Eberhardt, Thomas Icard, and Kara Schechtman, I developed a framework that gives us the best of both worlds: we explicitly model attributes like gender as social kinds, while preserving a precise formal model of discrimination. The key is to model protected attributes like gender as *high-level abstractions* of their constituents: by explicitly representing both causal and constitutive relationships, we model claims about discrimination as generalizations about the effects of different *ways* of varying gender (**Modeling Discrimination with Causal Abstraction**, R&R, *Philosophers' Imprint*). Our work was nominated for the Best Paper Award at the 2026 ACM Conference on Fairness, Accountability, and Transparency. With Jane Castleman, Kara Schechtman, and Lydia Liu, I am using our framework to design a causal test for discrimination in large language models (LLMs). We use techniques from mechanistic interpretability to learn how a given LLM represents gender. We then detect discrimination by testing whether the model's decisions are influenced by features that are constitutive of gender, by the model's own lights. Unlike many other tests, our approach is not limited by the experimenter's ability to anticipate how the model might discriminate.

5. Artificial Intelligence and Structural Injustice

My research on the ethics of AI, which spans explainable AI, algorithmic fairness, and AI alignment, concerns *structural injustice*: how several decisions, which appear innocent in isolation, can compose to create a system that is unjust.

- My work in explainable AI is concerned with how, even if (say) race and gender do not influence a decision when taken in isolation, an algorithm can still discriminate on the basis of race or gender, taken together with other factors. With Harry Sha and Li-Yang Tan, I developed a method for producing explanations of automated decision making (**A Generalization of the Satisfiability Coding Lemma and Its Applications**, *International Conference on Theory and Applications of Satisfiability Testing*, 2022), which resolves a problem posed by Quine over 70 years ago and won the conference's Best Theory Paper Award.
- In a paper published in the premier algorithmic fairness venue (**Multiplicative Metric Fairness Under Composition**, *Foundations of Responsible Computing*, 2023), I show how different definitions of "similar treatment for similar individuals" can differentially determine whether *groups* of individuals are treated unfairly, and whether several decision-makers that are fair in isolation from each other can *compose into a system* which is unfair. I prove that the most popular definition of equality of opportunity for individuals fails to protect against structural injustice, whereas a definition that has been largely overlooked does better.
- Leading methods for alignment like Reinforcement Learning from Human Feedback train AI to align with individuals' preferences, but doing so implicitly aggregates preferences in ways that can violate basic principles of fairness. With several co-authors, I proposed a framework for using social choice to align machine learning models' choices with human values (**Social Choice for AI Alignment**, *International Conference on Machine Learning*, 2024), which was the third most-cited paper in social choice in 2024. My interest in social choice extends beyond AI: in a paper with Wesley Holliday, Chase Norman, Eric Pacuit, and Cynthia Wang (**Stable Voting and the Splitting of Cycles**, *Association for the Advancement of Artificial Intelligence*, 2026), we resolve a conjecture about the relationship between two major voting rules. I build on my work on social choice and alignment in an invited contribution to *Philosophy & Technology*, where I raise a number of ethical problems with industry-standard alignment methods (**Why AI Alignment Needs Moral Foundations**).

I have also published two papers on artificial intelligence in leading computer science venues. **Conditional Negative Sampling for Contrastive Learning of Visual Representations** (*International Conference on Machine Learning*, 2021) explains why certain transformations on datasets (e.g. adding random noise) improve learning, while others (e.g. rotating an image) do not. **Zero-Shot Learning for Code Education** (*Association for the Advancement of Artificial Intelligence*, 2019) provides a method for "zero-shot learning"—that is, learning without access to any examples. We use a fixed probabilistic grammar to generate samples, and the learner uses this simulated dataset instead of a real one. This paper won the conference's Outstanding Student Paper Award (presented to 1 out of over 7,095 submitted papers).

6. How Hard Are Probabilistic and Causal Reasoning?

Causal reasoning seems indisputably harder than "correlational" or probabilistic reasoning: it requires that we perform deliberate experimentation and that we accept strong causal assumptions. Even so, Duligur Ibeling, Thomas Icard, and I show that in a precise sense, causal and probabilistic reasoning are equally difficult. We show how to computationally reduce problems about causation to equivalent ones about correlation: the computational complexity of deductive probabilistic and causal reasoning is the same (**Is Causal Reasoning Harder than Probabilistic Reasoning?**, *Review of Symbolic Logic*, 2022). Bläser, Dörfler, Liśkiewicz, and van der Zander extend our systems to include a summation operator, a feature of Pearl's influential do-calculus for causation. In reply, we show that suitably restricted, the operator does not change our results, but that when the operator is interpreted in the standard way, the do-calculus is undecidable. Positively, our work bounds the complexity of popular causal inference tools and shows that causality adds no computational cost. Negatively, we show there is no procedure for causal reasoning, as it is traditionally modeled (**On Probabilistic and Causal Reasoning with Summation Operators**, *Journal of Logic and Computation*, 2024). Logicians and psychologists often rely on an elusive distinction between *qualitative* and *quantitative* probabilistic reasoning. Duligur Ibeling, Thomas Icard, Krzysztof Mierzewski, and I provide a robust characterization of the distinction using computational complexity (**Probing the Qualitative-Quantitative Divide**, *Annals of Pure and Applied Logic*, 2023). In **Reasoning about Confirmation** (under review), which I presented at Carnegie Mellon's 2025 Probability Logic Workshop, I strengthen our prior results and along the way resolve some open questions about the difficulty of reasoning about when a piece of evidence confirms a hypothesis. I am interested in related topics in bounded rationality and cognitive science, e.g. parameterized complexity theory and the difficulty of reasoning about probabilistic independence. I hope to pursue these issues in future collaborations.

7. Teleological Explanations

Our descriptions of mental states, obligations, and the functions of artifacts and living things are often *teleological*, featuring sequences of means-end relations:

(1) John plans to go to the store to get thread to fix his jacket.
(2) We should compost to reduce methane emissions to slow global warming.
(3) The dishwasher ran its motor to splash suds to clean the dishes.
(4) The seedling needs water to grow to reach the sunlight.

The explanations differ markedly, but they share a common structure. We now know *why* John plans to go to the store, and *how* he plans to fix his jacket, just as we know *why* the dishwasher runs its motor and *how* it cleans the dishes. Taking these data as our cue, Yalcin and I developed a formal model of means-end relations using how- and why-questions. We show that this leads to a new model of planning states of mind (**How Questions Structure Plans**), while furnishing novel semantics for a substantial swath of practical and normative language, including deontic modals and need, desire, hope, and other attitude ascriptions. In 2026, we presented our work at the Eastern APA, the Planning Workshop at Stanford, and the annual meeting of the Philosophy of Language Association at UCLA.